Three-dimensional optical positioning using a compact inclined PD array with joint position-attitude estimation

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Abstract: This paper addresses the position–attitude coupling problem inherent to single light-emitting diode (LED) visible light positioning (VLP) under free receiver orientations. A compact inclined photodetector (PD) array integrated with an inertial measurement unit (IMU) is proposed to jointly extract received signal strength (RSS), two-dimensional angle of arrival (AOA), and attitude prior information. Based on these measurements, a joint position–attitude estimation algorithm is developed that couples optical measurement residuals with IMU constraints, enabling iterative refinement of position and attitude while limiting the propagation of attitude errors. Experimental results demonstrate average positioning errors of approximately 7 cm on a fixed receiving plane and 15 cm in three-dimensional space under free-tilt conditions, confirming the feasibility of accurate single-LED positioning in practical mobile scenarios.

1. Introduction

The constraints of traditional radio-frequency (RF) methods and the increasing need for indoor localization have prompted advancements in visible light communication and positioning (VLPC) technologies. Utilizing current lighting infrastructure and the directional properties of optical transmission, VLPC systems provide benefits in scalable deployment and precise positioning. However, practical deployment necessitates reliable performance across a range of receiver positions and orientations.

In response to these application requirements, prior research on indoor visible-light positioning and communication has focused on positioning algorithms, receiver architectures, and multisensor fusion. Zhu et al. introduced an angle-difference-of-arrival (ADOA)-based approach for three-dimensional (3D) positioning with arbitrarily tilted receivers, achieving an average experimental error of approximately 3.20 cm [1]. Kim et al. developed a 3D received signal strength (RSS) channel model that incorporates transmitter–receiver geometry and nonlinear optimization, resulting in centimeter-level positioning accuracy within the experimental region [2]. Bastiaens et al. enhanced RSS-based positioning by more accurately modeling the incidence-angle response of photodiodes, thereby reducing errors associated with the conventional Lambertian assumption [3]. Imajuku et al. presented an angle-of-arrival (AOA) estimation technique utilizing a photodetector array and a multiple-input multiple-output (MIMO) processing framework for source separation and angular estimation [4]. Aparicio-Esteve et al. leveraged geometric constraints from multiple receivers and nonlinear least-squares optimization to achieve high-accuracy 3D positioning under redundant observation conditions [5]. Ma et al. integrated multiple photodetectors (PDs) with deep learning, demonstrating a practical prototype capable of centimeter-level and sub-centimeter-level positioning accuracy [6]. Sun et al. combined visible light positioning (VLP) with an inertial measurement unit (IMU) using graph optimization, improving robustness to signal blockage and variations in receiver orientation [7].

To ensure the solvability of single-LED positioning, existing VLP studies commonly simplify receiver orientation [4–11,21,22,25], reduce 3D positioning to an equivalent 2D

problem under fixed-height assumptions [4,7,8,10,11,20], or introduce additional optical constraints using multiple LEDs or spatially separated receivers [1–5,9,12,20,23–25]. Although these strategies improve problem conditioning, they impose orientation, height, or hardware constraints. Moreover, the propagation of orientation errors through nonlinear observation models has received limited attention, and some studies employ relatively restricted experimental ranges [2–6,8,13,20–25]. This study further examines how attitude priors influence the solvability and achievable accuracy of single-LED positioning with unconstrained receiver orientation, utilizing Fisher information and the Cramér–Rao lower bound (CRLB).

2. System model

2.1 Coordinate systems and geometric definitions

The spatial geometric relationship between a single LED and the receiver is defined by establishing a world Cartesian coordinate system $\{W\} \triangleq (O^W, X^W, Y^W, Z^W)$ and a receiver body coordinate system $\{B\} \triangleq (O^B, \mathbf{u}_{PD}^B, \mathbf{v}_{PD}^B, \mathbf{n}_{PD}^B)$, as shown in Fig. 1.

Fig. 1 defines the world coordinate system and the receiver body coordinate system, with the latter centered at the central PD. Unless otherwise specified, we express all quantities in the world coordinate system.

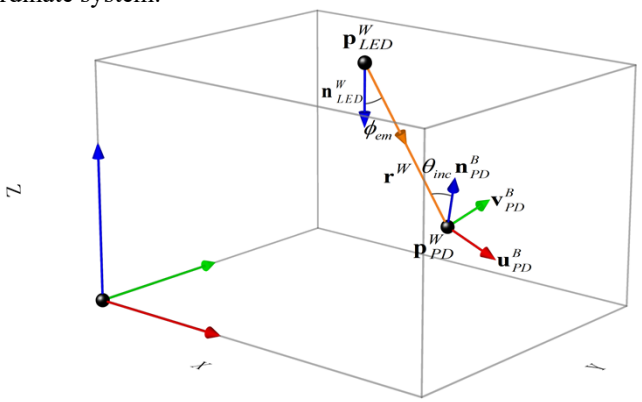


Fig. 1. Indoor geometry and coordinate systems.

The positions of the LED and the reference point of the photodetector (PD) array in the world coordinate system are denoted as $\mathbf{p}_{LED}^W = [x_{LED}, y_{LED}, z_{LED}]^{\mathrm{T}}$ and $\mathbf{p}_{PD}^W = [x_{PD}, y_{PD}, z_{PD}]^{\mathrm{T}}$, respectively. $\mathbf{r}^W = \mathbf{p}_{PD}^W - \mathbf{p}_{LED}^W = [r_x, r_y, r_z]^{\mathrm{T}}$ represents the propagation vector from the transmitter to the receiver, $d = \left\|\mathbf{r}^W\right\|_2$ the propagation distance, and $\hat{\mathbf{r}}^W = \mathbf{r}^W / d$ the corresponding unit propagation direction. The LED optical axis is assumed to be oriented vertically downward, resulting in $\mathbf{n}_{LED} = [0, 0, -1]^{\mathrm{T}}$.

The receiver body coordinate system is established using the orthonormal basis $\{\mathbf{u}_{PD}^B, \mathbf{v}_{PD}^B, \mathbf{n}_{PD}^B\}$, where $\mathbf{n}_{PD}^B$ aligns with the effective surface normal of the central PD. The receiver's orientation is characterized by the rotation matrix $\mathbf{R}_{WB}$, which transforms vectors from the body coordinate system to the world coordinate system. Consequently, the normal vector of the central PD, when expressed in the world coordinate system, is denoted as $\mathbf{n}_{PD}^W = \mathbf{R}_{WB}\mathbf{n}_{PD}^B$. The unit incident direction, defined as the vector pointing from the PD toward the LED, is represented as $\hat{\mathbf{k}}_{in}^W = -\hat{\mathbf{r}}^W$.

2.2 LOS radiation and optical channel model

$\Phi_{1/2}$ denotes the half-power semi-angle of the LED. The Lambertian order is represented by $m = -\ln 2 / \ln(\cos \Phi_{1/2})$. The irradiance angle from the LED to the photodetector (PD) is defined as $\phi_{em} = \arccos(\mathbf{n}_{LED}^{\mathrm{T}} \hat{\mathbf{r}})$, with $\phi_{em} \le 90^{\circ}$. The incidence angle at the PD is denoted by $\theta_{inc} = \arccos(\mathbf{n}_{PD}^{\mathrm{T}} \hat{\mathbf{k}}_{in}) = \arccos(-\mathbf{n}_{PD}^{\mathrm{T}} \hat{\mathbf{r}})$. The half field of view (FOV), represented by θ_{FOV}, specifies the maximum incidence angle that can be accepted by the PD. When $\theta_{inc} > \theta_{FOV}$ or $\mathbf{n}_{PD}^{\mathrm{T}} \hat{\mathbf{k}}_{in} \le 0$, meaning the incident direction is behind the PD receiving surface, the line-of-sight (LOS) channel gain is set to zero. Neglecting multipath reflections and scattering, the LOS optical channel gain is given by

$$H_{LOS}(d, \phi_{em}, \theta_{inc}) = \begin{cases} \dfrac{C_1}{d^2} \cos^m(\phi_{em}) \cos(\theta_{inc}), 0 < \theta_{inc} < \theta_{FOV} \text{ and } \cos\theta_{inc} > 0, \\ 0, \text{otherwise}, \end{cases} \tag{1}$$

where $C_1 = A_{PD} \eta_{opt} T_s (m+1) / 2\pi$, A_{PD} is the effective photosensitive area of the PD, η_{opt} denotes the optical-path efficiency factor, and T_s is the transmittance of the optical window or filter. Unless otherwise stated, $H_{LOS}(d, \phi_{em}, \theta_{inc})$ and the related expressions in this work are assumed to satisfy the FOV and irradiance-angle constraints. Let $\Re$ denote the responsivity of the PD and Φ the average radiant power of the LED. The signal-induced photocurrent is then given by

$$I_{sig} = \Re \Phi H_{LOS}(d, \phi_{em}, \theta_{inc}) = \frac{C_2}{d^2} \cos^m(\phi_{em}) \cos(\theta_{inc}) \quad , \tag{2}$$

where $C_2 = \Re \Phi C_1$. Taking the dark current and receiver noise into account, the measured photocurrent can be expressed as

$$I_{meas} = I_{sig} + I_{dark} + v_I \quad , \tag{3}$$

where $v_I \sim \mathbb{N}(0, \sigma_I^2)$ denotes zero-mean Gaussian noise with variance σ_I^2. Assuming that the individual noise sources are mutually independent, the total equivalent input-current noise variance can be written as

$$\sigma_I^2 = 2q(I_{\mathrm{sig}} + I_{dark}) B_e + (\Re \cdot NEP)^2 B_e + i_{read}^2 B_e \quad , \tag{4}$$

where $q = 1.602 \times 10^{-19} C$, B_e, NEP, and i_{read} denote the elementary charge, equivalent noise bandwidth, noise-equivalent power, and readout noise density, respectively. Their values are summarized in Table 1. Before positioning, the dark-current dc offset is removed through calibration. Therefore, the RSS measurement is given by

$$z_{RSS} = I_{meas} - I_{dark} \quad . \tag{5}$$

2.3 Coupling between position and orientation

According to the LOS channel model, the received photocurrent is determined not only by the propagation distance and LED irradiance angle, but also by the incidence angle at the PD. For a receiver with arbitrary orientation, the incidence angle satisfies

$$\cos\theta_{inc} = -\left(\mathbf{R}_{WB} \mathbf{n}_{PD}^{B}\right)^{\mathrm{T}} \hat{\mathbf{r}}^{W} \quad , \tag{6}$$

where the propagation direction $\hat{\mathbf{r}}^W$ depends on the receiver position, while the rotation matrix $\mathbf{R}_{WB}$ depends on the receiver orientation. Conventional positioning methods often assume a known receiver orientation to simplify position estimation; however, orientation errors may propagate into the position estimate through the incidence-angle response. In this work, position and orientation are jointly estimated to reduce such error propagation.

3. Compact inclined PD array and joint position–orientation estimation

3.1 Compact inclined PD array and 2D AOA formulation

To introduce additional geometric constraints without increasing the number of transmitter anchors, a compact inclined PD array is constructed by arranging N neighboring PDs around a central PD in the receiver body coordinate system $\{B\}$. The neighboring PDs are uniformly distributed on a circle with radius r, and their normal vectors are tilted outward along the radial direction by an angle γ with respect to the central PD normal vector $\mathbf{n}_{PD}^B$, forming $\mathbf{n}_j^B$. As illustrated in Fig. 2, the local position and normal vector of the central PD are defined as $\mathbf{p}_0^B = [0,0,0]^{\mathrm{T}}$ and $\mathbf{n}_0^B = [0,0,1]^{\mathrm{T}}$, respectively. Identical PD devices and receiver circuits are assumed for modeling, simulation, and experiments.

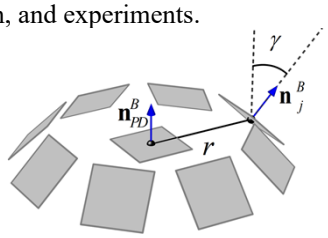


Fig. 2. Schematic of the compact inclined PD array.

The azimuth angle of the j th neighboring PD in the PD plane is

$$\theta_j = (j-1)\frac{2\pi}{N}, j = 1,...,N \quad . \tag{7}$$

Accordingly, its local position and normal vector are expressed as $\mathbf{p}_j^B = \left[r\cos\theta_j, r\sin\theta_j 0,\right]^{\mathrm{T}}$ and $\mathbf{n}_j^B = \left[\sin\gamma\cos\theta_j, \sin\gamma\sin\theta_j, \cos\gamma\right]^{\mathrm{T}}$, $\left\|\mathbf{n}_j^B\right\|_2 = 1$, respectively. Under $r \ll d$, distance-dependent attenuation differences among neighboring PDs can be neglected, and the inter-channel variations are mainly governed by the incidence-angle responses. The first-order spatial-baseline effect introduced by the array radius is retained and further analyzed in Sec. 3.3. At the kth instant, the dc photocurrent measured by the jth PD can be written as

$$z_{RSS,j,k} = I_{sig,j,k} + v_{I,j,k} = K_k \left(\mathbf{n}_j^B\right)^{\mathrm{T}} \hat{\mathbf{k}}_{in,k}^B + v_{I,j,k} \quad , \tag{8}$$

where $K_k = \left(C_2 / d_k^2\right)\cos^m(\phi_{em,k})$ collects the common amplitude-related terms of the array channels, including distance attenuation, Lambertian radiation, and system constants, and $v_{I,j,k}$ denotes the equivalent input current noise. For compact representation, let $\mathbf{d}_k = K_k \hat{\mathbf{k}}_{in,k}^B$, $\mathbf{N}=\left[(\mathbf{n}_0^B)^{\mathrm{T}},...,(\mathbf{n}_M^B)^{\mathrm{T}}\right]$, which gives the linear relationship between the photocurrent vector and the incident direction

$$\mathbf{z}_{\mathrm{RSS},k} = \left[z_{RSS,0k},...,z_{RSS,M,k}\right]^{\mathrm{T}} \approx \mathbf{N}\mathbf{d}_{\mathrm{k}} \quad , \tag{9}$$

where, $M \le N$ denotes the number of effective channels participating in the inversion. The incident direction is estimated by least squares and subsequently normalized as

$$\tilde{\mathbf{d}}_k = \left(\mathbf{N}^{\mathrm{T}}\mathbf{N}\right)^{-1}\mathbf{N}^{\mathrm{T}}\mathbf{z}_{\mathrm{RSS},k} \quad , \tag{10}$$

$$\hat{\mathbf{k}}_{in,k}^{B} = \frac{\tilde{\mathbf{d}}_{\mathrm{k}}}{\left\|\tilde{\mathbf{d}}_k\right\|} \quad . \tag{11}$$

Equation (10) shows that an excessively small inclination angle γ provides insufficient information on the horizontal incident direction, whereas an excessively large γ weakens the vertical-direction information. Therefore, γ determines the conditioning of the two-dimensional AOA reconstruction and the observability of the three-dimensional incident direction. The horizontal AOA α_k and vertical AOA β_k can then be obtained from the reconstructed unit direction vector. The horizontal AOA is defined as the angle between the projection of the incident direction onto the PD plane and the $\mathbf{u}_{PD}^{B}$ axis, whereas the vertical AOA corresponds to the incidence angle θ_{inc}. Their estimates are given by $\tilde{\alpha}_k = \mathrm{atan2}(\hat{k}_{y,k}^{\mathrm{B}}, \hat{k}_{x,k}^{\mathrm{B}})$, $\tilde{\beta}_k = \arccos(\hat{k}_{z,k}^{\mathrm{B}})$.

A unique reconstruction of the incident direction from (10) requires $\mathbf{N}^{\mathrm{T}}\mathbf{N}$ to be invertible. For a uniformly distributed neighboring-PD set with $M \ge 3$, independent directional constraints are provided and $\mathbf{N}^{\mathrm{T}}\mathbf{N}$ is full rank, with $\det\left(\mathbf{N}^{\mathrm{T}}\mathbf{N}\right) = \left(M^2/4\right)\sin^4\gamma\left(1 + M\cos^2\gamma\right) \ne 0$. Hence, the three-dimensional incident direction can be uniquely reconstructed in the receiver coordinate system and subsequently mapped to the world coordinate system using the IMU attitude information. Combined with RSS-based distance information, the AOA measurements provide sufficient geometric constraints for single-LED positioning under free-attitude conditions.

3.2 Position–attitude joint estimation model under attitude-prior constraints

Under free-attitude conditions, attitude errors can propagate into the photocurrent and AOA measurements, thereby degrading position estimation. Two estimation modes are considered: Mode A fixes the IMU attitude and estimates only the receiver position, whereas Mode B jointly estimates position and attitude using the IMU attitude as a prior.

The theoretical RSS-AOA observation function is written as

$$\mathbf{h}\left(\mathbf{p}_{PD,k}, \mathbf{q}_{WB,k}\right) = \begin{bmatrix} K_k\left(\mathbf{n}_0^{B}\right)^{\mathrm{T}}\hat{\mathbf{k}}_{in,k}^{B} \\ \cdots \\ K_k\left(\mathbf{n}_M^{B}\right)^{\mathrm{T}}\hat{\mathbf{k}}_{in,k}^{B} \\ \mathrm{atan2}(\hat{k}_{y,k}^{\mathrm{B}}, \hat{k}_{x,k}^{\mathrm{B}}) \\ \arccos(\hat{k}_{z,k}^{\mathrm{B}}) \end{bmatrix} \quad , \tag{12}$$

where $\mathbf{q}_{WB,k}$ denotes the receiver attitude quaternion at the k th instant, and the corresponding rotation matrix is $\mathbf{R}_{WB,k}$. The positioning observation model is then given by

$$\mathbf{z}_{pos,k} = \left[\mathbf{z}_{\mathrm{RSS},k}, \tilde{\alpha}_k, \tilde{\beta}_k\right]^{\mathrm{T}} = \mathbf{h}\left(\mathbf{p}_{PD,k}, \mathbf{q}_{WB,k}\right) + \mathbf{v}_k \quad , \tag{13}$$

where $\mathbf{v}_k = \left[\mathbf{v}_{RSS,k}, v_{\alpha,k}, v_{\beta,k}\right]^{\mathrm{T}} \sim \mathbb{N}\left(0, \mathbf{R}_k\right)$ denotes the Gaussian observation noise, and $\mathbf{R}_k = \mathrm{diag}\left(\sigma_I^2 \mathbf{E}_{M+1}, \sigma_{\alpha,k}^2, \sigma_{\beta,k}^2\right)$ is the corresponding noise covariance matrix. The observation residual between the actual measurement and the theoretical prediction is defined as

$$\mathbf{e}_k(\mathbf{p}_{PD,k}, \mathbf{q}_{WB,k}) = \mathbf{z}_{pos,k} - \mathbf{h}(\mathbf{p}_{PD,k}, \mathbf{q}_{WB,k}) \quad . \tag{14}$$

Let $\tilde{\mathbf{q}}_{WB,k}$ denote the prior attitude quaternion provided by the IMU attitude-estimation module at the kth instant, representing the rotation from the receiver body coordinate system to the world coordinate system. The error quaternion between the current attitude estimate and the attitude prior is approximated by

$$\delta\mathbf{q}_k \approx \left[1, \frac{1}{2}\delta\boldsymbol{\theta}_k\right]^{\mathrm{T}} \quad , \tag{15}$$

where $\delta\boldsymbol{\theta}_k \sim \mathbb{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\theta,k})$, and $\boldsymbol{\Sigma}_{\theta,k} \approx \mathrm{diag}(\sigma_{roll}^2, \sigma_{pitch}^2, \sigma_{yaw}^2)$ denotes the covariance matrix of the IMU attitude-prior error (roll, pitch, and yaw).

In Mode A, the IMU attitude is fixed and only the receiver position is estimated. To account for attitude-prior uncertainty, the attitude Jacobian is defined as

$$\mathbf{H}_{\theta,k} = \frac{\partial \mathbf{h}}{\partial \delta\boldsymbol{\theta}_k} \quad . \tag{16}$$

Under the small-attitude-error assumption, first-order Taylor expansion and error propagation show that the attitude-prior uncertainty introduces an additional covariance term into the observation model. Accordingly, the effective observation-noise covariance and the position estimate are given by

$$\mathbf{R}_{eff,k} = \mathbf{R}_k + \mathbf{H}_{\theta,k} \boldsymbol{\Sigma}_{\theta,k} \mathbf{H}_{\theta,k}^{\mathrm{T}} \quad , \tag{17}$$

$$\tilde{\mathbf{p}}_{PD,k} = \arg\ \min\ \mathbf{e}_{A,k}^{\mathrm{T}} \mathbf{R}_{eff,k}^{-1} \mathbf{e}_{A,k} \quad , \tag{18}$$

where $\mathbf{e}_{A,k} = \mathbf{z}_{pos,k} - \mathbf{h}(\mathbf{p}_{PD,k}, \tilde{\mathbf{q}}_{WB,k})$. In Mode B, the position and attitude are jointly treated as unknown variables. The attitude-prior residual and the joint position-attitude estimate are defined as

$$\mathbf{e}_{\theta,k} = 2\ \mathrm{vec}\left[(\tilde{\mathbf{q}}_{WB,k})^{-1} \otimes \mathbf{q}_{WB,k}\right] \approx \delta\boldsymbol{\theta}_k \quad , \tag{19}$$

$$(\tilde{\mathbf{p}}_{PD,k}, \tilde{\mathbf{q}}_{WB,k}) = \arg\ \min\ \left(\mathbf{e}_k^{\mathrm{T}} \mathbf{R}_k^{-1} \mathbf{e}_k + \mathbf{e}_{\theta,k}^{\mathrm{T}} \boldsymbol{\Sigma}_{\theta,k}^{-1} \mathbf{e}_{\theta,k}\right) \quad , \tag{20}$$

where $\otimes$ denotes quaternion multiplication, and $\mathrm{vec}(\cdot)$ extracts the three-dimensional vector part of a quaternion. Mode A absorbs the attitude uncertainty into the observation noise, whereas Mode B explicitly corrects the attitude using the optical observations.

3.3 Cramér–Rao lower-bound analysis of joint estimation

(1) Position and attitude observation Jacobians

To derive the explicit position Jacobian of the jth RSS observation, define the projection matrix orthogonal to the LOS direction as $\mathbf{P}_k = \mathbf{E}_3 - \hat{\mathbf{r}}_k \hat{\mathbf{r}}_k^{\mathrm{T}}$. From the derivatives of K_k and the unit incident direction, we obtain

$$\frac{\partial K_k}{\partial \mathbf{p}_{PD,k}} = \frac{K_k}{d_k}\left[-2\hat{\mathbf{r}}_k^{\mathrm{T}} + \frac{m}{\mathbf{n}_{LED}^{\mathrm{T}}\hat{\mathbf{r}}_k}\mathbf{n}_{LED}^{\mathrm{T}}\mathbf{P}_k \right] , \tag{21}$$

$$\frac{\partial \mathbf{k}_{in,k}^{B}}{\partial \mathbf{p}_{PD,k}} = -\frac{\mathbf{R}_{WB,k}^{\mathrm{T}}\mathbf{P}_k}{d_k} . \tag{22}$$

Therefore, the position Jacobian of the j th RSS observation is

$$\mathbf{H}_{I_j,p,k} = \left[\left(\mathbf{n}_j^{B}\right)^{\mathrm{T}} \mathbf{k}_{in,k}^{B} \right] \frac{\partial K_k}{\partial \mathbf{p}_{PD,k}} - \frac{K_k}{d_k}\left(\mathbf{n}_j^{B}\right)^{\mathrm{T}} \mathbf{R}_{WB,k}^{\mathrm{T}}\mathbf{P}_k , \tag{23}$$

the first term accounts for the effects of distance attenuation and variation in the LED irradiance angle, whereas the second term represents the change in the incident direction.

The Jacobian of the incident direction with respect to the attitude error $\delta\boldsymbol{\theta}_k$ is

$$\mathbf{J}_{k,\theta} = \frac{\partial \mathbf{k}_{in,k}^{B}}{\partial \delta\boldsymbol{\theta}_k} . \tag{24}$$

Accordingly, the attitude Jacobian of the jth RSS observation is

$$\mathbf{H}_{I_j,\theta,k} = K_k \left(\mathbf{n}_j^{B}\right)^{\mathrm{T}} \mathbf{J}_{k,\theta} . \tag{25}$$

The Jacobian of the AOA with respect to the incident direction is

$$\mathbf{G}_k = \frac{\partial \left[\alpha_k, \beta_k\right]^{\mathrm{T}}}{\partial \mathbf{k}_{in,k}^{B}} . \tag{26}$$

By applying the chain rule, the Jacobians of the two-dimensional AOA with respect to the position and attitude are obtained as

$$\mathbf{H}_{AOA,p,k} = -\frac{\mathbf{G}_k\mathbf{R}_{WB,k}^{\mathrm{T}}\mathbf{P}_k}{d_k} , \tag{27}$$

$$\mathbf{H}_{AOA,\theta,k} = \mathbf{G}_k\mathbf{J}_{k,\theta} . \tag{28}$$

By stacking the RSS and AOA Jacobians according to the observation order in Eq. (13), the position Jacobian of the observation function is obtained as

$$\mathbf{H}_{p,k} = \frac{\partial \mathbf{h}}{\partial \mathbf{p}_{PD,k}} . \tag{29}$$

(2) Position error bounds of Modes A and B

For Mode A, the position Fisher information matrix and the position error bound (PEB) are given by

$$\mathbf{J}_{A,k} = \mathbf{H}_{p,k}^{\mathrm{T}}\mathbf{R}_{eff,k}^{-1}\mathbf{H}_{p,k} , \tag{30}$$

$$\mathrm{PEB}_{A,k} = \sqrt{\mathrm{tr}\left(\mathbf{J}_{A,k}^{-1}\right)} . \tag{31}$$

For Mode B, the Fisher information associated with the position, position–attitude coupling, and attitude is respectively defined as

$$\mathbf{J}_{pp,k} = \mathbf{H}_{p,k}^{\mathrm{T}} \mathbf{R}_k^{-1} \mathbf{H}_{p,k} \quad , \tag{32}$$

$$\mathbf{J}_{p\theta,k} = \mathbf{H}_{p,k}^{\mathrm{T}} \mathbf{R}_k^{-1} \mathbf{H}_{\theta,k} \quad , \tag{33}$$

$$\mathbf{J}_{\theta\theta,k} = \mathbf{H}_{\theta,k}^{\mathrm{T}} \mathbf{R}_k^{-1} \mathbf{H}_{\theta,k} + \mathbf{\Sigma}_{\theta,k}^{-1} \quad . \tag{34}$$

These three terms represent the position information, position–attitude coupling information, and attitude information, respectively. Accordingly, the joint Fisher information matrix of Mode B is written as

$$\mathbf{J}_{B,k} = \begin{bmatrix} \mathbf{J}_{pp,k} & \mathbf{J}_{p\theta,k} \\ \mathbf{J}_{p\theta,k}^{\mathrm{T}} & \mathbf{J}_{\theta\theta,k} \end{bmatrix} \quad . \tag{35}$$

According to the block-matrix inversion relation, the position covariance bound corresponds to the upper-left block of $\mathbf{J}_{B,k}^{-1}$. Hence, the equivalent position FIM and PEB for Mode B are given by

$$\mathbf{J}_{p,B,k} = \mathbf{J}_{pp,k} - \mathbf{J}_{p\theta,k} \mathbf{J}_{\theta\theta,k}^{-1} \mathbf{J}_{p\theta,k}^{\mathrm{T}} \quad , \tag{36}$$

$$\mathrm{PEB}_{B,k} = \sqrt{tr\left(\mathbf{J}_{p,B,k}^{-1}\right)} \quad . \tag{37}$$

The second term in Eq. (36) represents the loss of position information induced by position–attitude coupling. In addition to this local information coupling, a finite deviation between the IMU attitude prior and the true attitude is further propagated to the RSS and AOA observations through the nonlinear observation model. The resulting observation error due to the attitude deviation is defined as

$$\Delta\mathbf{z}_{\theta,k} = \mathbf{h}(\mathbf{p}_{PD,k}, \mathbf{q}_{WB,k}) - \mathbf{h}(\mathbf{p}_{PD,k}, \tilde{\mathbf{q}}_{WB,k}) \quad , \tag{38}$$

where $\Delta\mathbf{z}_{\theta,k}$ denotes the observation error induced by the attitude deviation in the RSS and AOA measurements.When the attitude deviation is small, a first-order Taylor expansion yields the aforementioned error-propagation relationship,

$$\Delta\mathbf{z}_{\theta,k} \approx \mathbf{H}_{\theta,k} \delta\boldsymbol{\theta}_k \quad . \tag{39}$$

As the initial attitude deviation increases, higher-order nonlinear effects become non-negligible, leading to additional model mismatch in Mode A due to its fixed IMU attitude.

The IMU attitude prior provides additional attitude information through $\mathbf{\Sigma}_{\theta,k}^{-1}$, reducing the position–attitude coupling loss in Eq. (36). By applying the matrix inversion lemma to the effective covariance of Mode A, its equivalent position FIM becomes $\mathbf{J}_{A,k} = \mathbf{J}_{pp,k} - \mathbf{J}_{p\theta,k} \mathbf{J}_{\theta\theta,k}^{-1} \mathbf{J}_{p\theta,k}^{\mathrm{T}} = \mathbf{J}_{p,B,k}$. Therefore, under first-order linearization, Gaussian-error assumptions, and identical attitude priors, Modes A and B yield the same theoretical position bound. For large initial attitude deviations, Mode B further refines the attitude through joint estimation, thereby improving robustness against attitude-error propagation.

(3) Effects of array inclination angle and radius on PEB

Under the aforementioned approximation $r \ll d$, differences in distance attenuation and LED irradiance angle among neighboring PDs are neglected, while the first-order spatial-baseline differences introduced by the array radius are retained. Define the radial unit vector of the jth neighboring PD in the array plane as $\mathbf{u}_j = \left[\cos\theta_j, \sin\theta_j, 0\right]^{\mathrm{T}}$, so that its local position relative

to the central PD is $\mathbf{p}_j^B = r\mathbf{u}_j$. Let $\hat{\mathbf{k}}_{0,in,k}^B$ denote the unit direction from the central PD toward the LED. The unit incident direction of the jth neighboring PD can then be approximated to first order as

$$\hat{\mathbf{k}}_{j,in,k}^B \approx \hat{\mathbf{k}}_{0,in,k}^B - \frac{r}{d_k}\left[\mathbf{E}_3 - \hat{\mathbf{k}}_{0,in,k}^B \left(\hat{\mathbf{k}}_{0,in,k}^B\right)^{\mathrm{T}}\right]\mathbf{u}_j \quad . \tag{40}$$

For the representative near-normal incidence condition $\hat{\mathbf{k}}_{0,in,k}^B = [0,0,1]^{\mathrm{T}}$, the symmetry of the uniformly distributed circular array can be exploited to sum the Fisher information contributed by all channels. The resulting local position FIM for given array parameters can be approximated as

$$\mathbf{J}_{r\gamma,k} \approx \frac{NK_k^2}{\sigma_I^2 d_k^2}\ \mathrm{diag}\left[\frac{1}{2}(\sin\gamma + \frac{r}{d_k}\cos\gamma)^2, \frac{1}{2}(\sin\gamma + \frac{r}{d_k}\cos\gamma)^2, (\frac{r}{d_k})^2 \sin^2\gamma\right] \quad . \tag{41}$$

The above expression explicitly reveals the effects of r and γ on the Fisher information. When $r \ll d$, increasing r improves the position Fisher information through a larger spatial baseline, whereas γ balances the directional information in different axes. For given array parameters γ and r, the average position error bound over N_s test states is defined as

$$\overline{\mathrm{PEB}}(\gamma,r) = \frac{1}{N_s}\sum_{i=1}^{N_s} \mathrm{PEB}(\mathbf{p}_{PD,i}, \mathbf{q}_{WB,i}; \gamma, r) \quad . \tag{42}$$

Therefore, the theoretical design of the compact inclined PD array can be obtained by scanning γ and r and minimizing Eq. (42).

4. Simulation and experimental results and analysis

The simulation and experimental studies employ consistent coordinate definitions, array configurations, and estimation algorithms. Only line-of-sight propagation is considered, while blockage and higher-order reflections are neglected.

4.1 Simulation results

Unless otherwise specified, the simulation parameters are listed in Table 1. Monte Carlo simulations are performed over receiver position, attitude, and noise, with identical initial conditions for Modes A and B.The receiver position is randomly generated within $(0.1 \sim L_x - 0.1, 0.1 \sim L_y - 0.1, 0.1 \sim L_z - 0.5)$, with pitch and roll in [-45°,45°] and yaw in [0°,360°). For two-dimensional positioning, the receiving-plane reference point is set to $\mathbf{p}_\Pi = [2.5, 2.0, 0.8]^{\mathrm{T}}$.

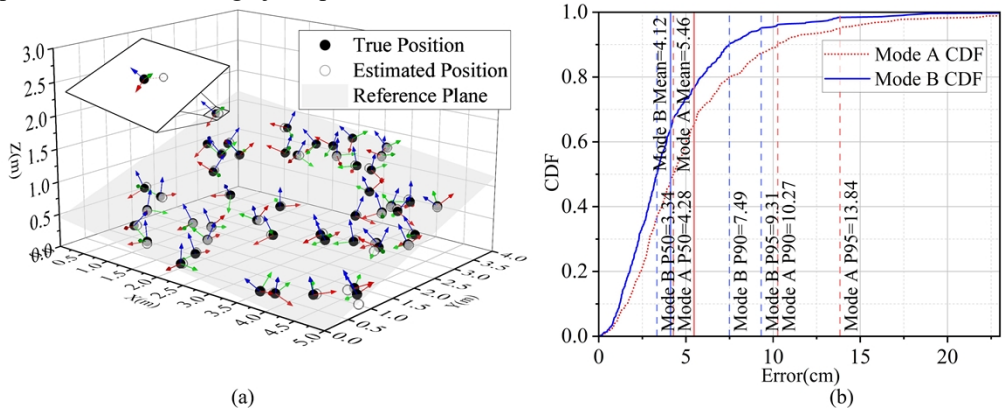
Table 1. Simulation parameter settings

Parameter	Value	Parameter	Value
Simulation region (L_x, L_y, L_z) (m)	5×4×3	Number of neighboring PDs N	8
LED position $\mathbf{p}_{LED}^W$ (m)	[2.5,2,3]^{T}	PD active area A_{PD} mm^2	0.49
LED normal $\mathbf{n}_{LED}^W$ W	[0,0,-1]^{T}	PD responsivity $\Re$ (A/W)	0.35
Average LED radiant power Φ (W)	2.1	PD window transmittance T_s	0.95

LED half-power semi-angle $\Phi_{1/2}$ (°)	60	Optical-link efficiency η_{opt}	0.85
Lambertian order m	1	PD field-of-view semi-angle θ_{FOV} (°)	90
Array radius r (m)	0.02	PD noise power NEP (pW / $\sqrt{\text{Hz}}$)	0.033
Inclination angle of neighboring PDs γ (°)	45	Readout-noise density σ_{read} (A/$\sqrt{\text{Hz}}$)	0.6×10^{-15}

(1) Two-dimensional positioning performance

For the fixed receiving-plane case, $\mathbf{n}_{\Pi}$ is normalized from $[0.1,-0.3,1]^{\mathrm{T}}$. The simulation results are shown in Fig. 3, where Fig. 3(a) presents the spatial scatter plot and Fig. 3(b) the cumulative distribution function (CDF) of positioning error. The 50th-percentile positioning errors of Modes A and B are 4.28 and 3.34 cm, respectively, while the corresponding 90th-percentile errors are 10.27 and 7.49 cm. The mean errors are 5.46 and 4.12 cm, respectively. These results indicate that Mode B achieves improved positioning accuracy under attitude-prior deviations through joint position – attitude refinement.

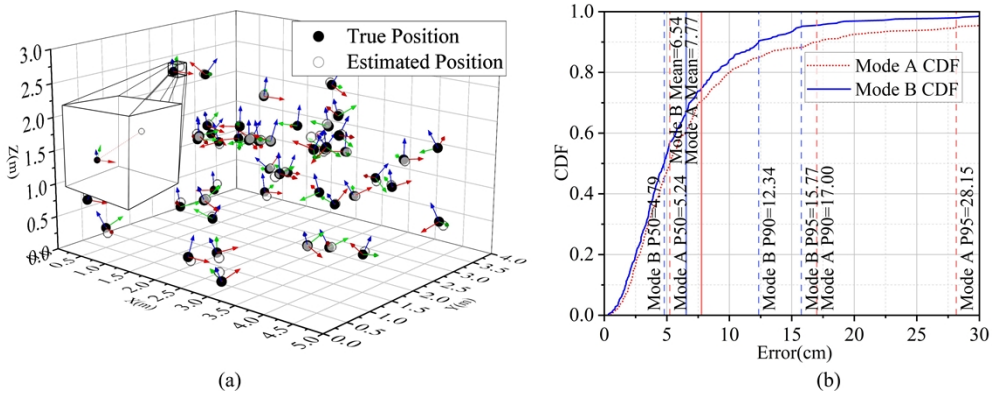


(a) (b)

Fig. 3. Two-dimensional positioning simulation results. (a) Spatial scatter plot. (b) CDF of positioning error.

(2) Three-dimensional positioning performance

In the three-dimensional free-space scenario, both the receiver position and attitude vary freely. Figure 4 presents the three-dimensional positioning results.



(a) (b)

Fig. 4. Three-dimensional positioning simulation results. (a) Spatial scatter plot. (b) CDF of positioning error.

The 50th-percentile positioning errors of Modes A and B are 5.24 and 4.79 cm, respectively, while the corresponding 90th-percentile errors are 17 and 12.34 cm. The mean errors are 7.77 and 6.54 cm, respectively. These results indicate that Mode B reduces the propagation of attitude deviations into the position estimate through joint position – attitude optimization, thereby improving overall positioning accuracy over Mode A.

(3) Analysis of factors affecting positioning error

The pitch, roll, and yaw angles are all set to 0°. For the fixed-plane scenario, $\mathbf{n}_{\Pi} = [0,0,1]^{\mathrm{T}}$ is imposed, and Mode B is adopted for positioning. The spatial distribution of the positioning error is presented in Fig. 5. As the receiver moves away from the LED projection center, the RSS and AOA observations become less accurate, leading to an overall increase in positioning error toward the boundary. A local high-error region is also observed near the center, mainly due to the reduced directional sensitivity of the AOA measurements.

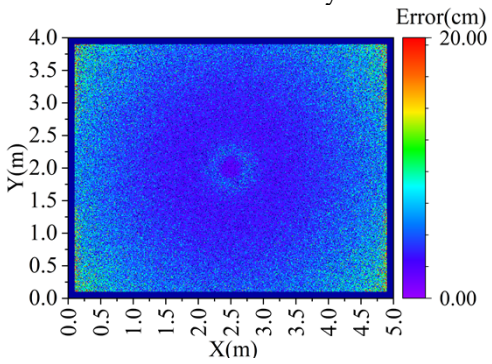


Fig. 5. Simulation results for the effect of receiver position on positioning performance.

The pitch and roll angles are randomly varied, while the yaw angle is fixed at 0°. For the fixed-plane scenario, $\mathbf{n}_{\Pi} = [0,0,1]^{\mathrm{T}}$, receiver position is fixed at $[2.5,2.0,0.8]^{\mathrm{T}}$. The simulation results of the two modes under attitude variations are shown in Fig. 6. Mode B achieves noticeably higher positioning accuracy than Mode A because it jointly optimizes the position and attitude, allowing part of the residual error to be compensated through attitude correction and thereby reducing the propagation of attitude errors into the position estimate.

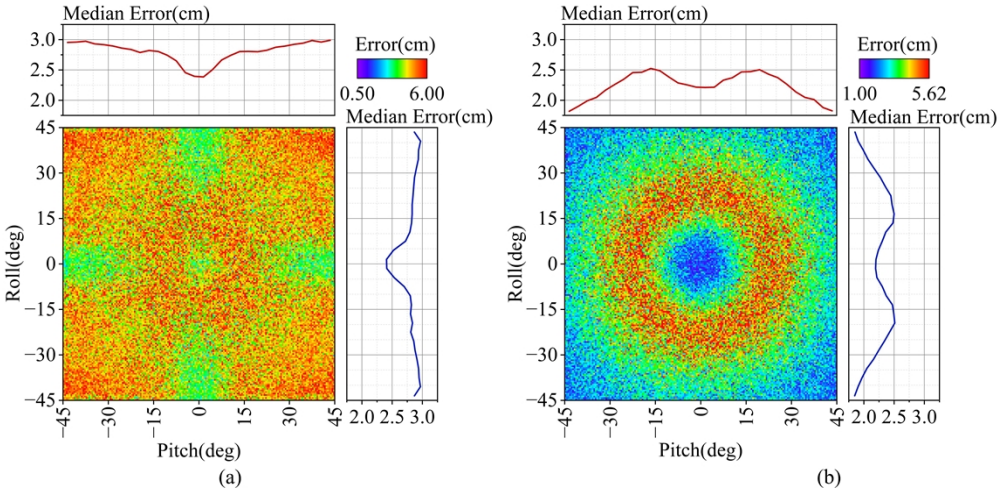


Fig. 6. Simulation results for the effect of receiver attitude on positioning performance. (a) Mode A. (b) Mode B.

Because AOA reconstruction depends on the dimensionality of the linearized model, its performance is affected by the number of effective channels M involved in the inversion. The pitch, roll, and yaw angles are randomly varied. For the fixed-plane scenario, $\mathbf{n}_{\Pi} = [0,0,1]^{\mathrm{T}}$ is imposed, and Mode B is adopted. The results are presented in Fig. 7.

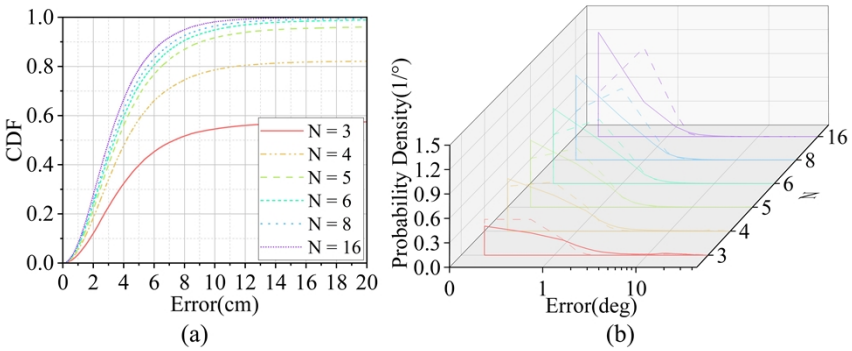


Fig. 7. Simulation results for the effect of the number of neighboring PDs on positioning performance. (a) CDF of positioning error. (b) Probability-density distributions of AOA angular errors.

In Fig. 7(b), the solid and dashed curves represent the probability densities of the horizontal and vertical AOA errors, respectively. As the number of PDs N increases, the observation redundancy is enhanced, resulting in significant reductions in both positioning and AOA errors. The improvement becomes gradual for $N > 6$, whereas the errors increase rapidly for $N < 6$. The broader distribution of the vertical AOA error indicates better angular resolution in the horizontal direction.

The pitch, roll, and yaw angles are randomly varied, and Mode B is adopted for three-dimensional positioning. The inclination angle γ is scanned from 0° to 90° in 1° increments, while the array radius r is varied from 0.5 to 20 cm in 0.5 cm increments. The simulated mean PEB and valid-sample ratio are presented in Fig. 8, where the contour lines denote the percentage of samples satisfying the LOS positioning condition. As γ increases, the directional discrimination of the array is enhanced, resulting in a marked reduction in the mean PEB over approximately 40°–60°. Increasing r further improves the spatial-baseline information, although an excessively large radius compromises array compactness. With γ =45° and r =2 cm, the mean PEB is approximately 5.25 cm and the valid-sample ratio reaches 99.7%, providing a favorable trade-off among positioning accuracy, sample validity, and structural compactness.

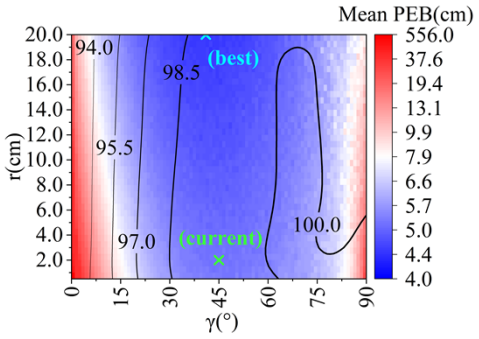
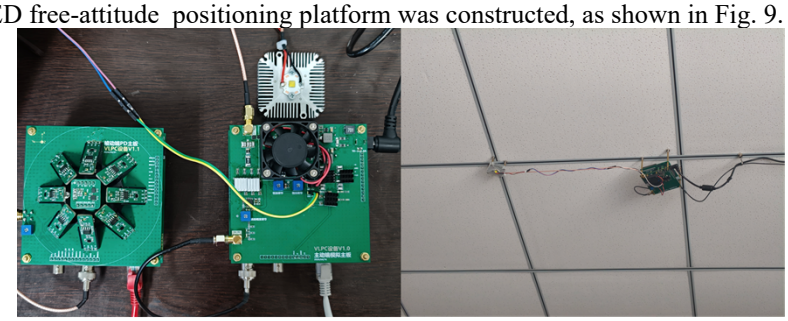


Fig. 8. Effect of array inclination angle and radius on the average PEB.

4.2 Experimental results

(1) Experimental platform and parameters

To validate the feasibility of the proposed method under practical nonideal conditions, a single-LED free-attitude positioning platform was constructed, as shown in Fig. 9.



(a) (b)

Fig. 9. Experimental platform. (a) Experimental prototype. (b) Experimental environment.

The experimental space measures 7.05 m × 6.37 m × 3.2 m, and the LED is located at $\mathbf{p}_{LED}^{W}=\left[3.72,3.72,3.16\right]^{\mathrm{T}}$ .The average radiant power of the LED is approximately $\Phi=12.5$ W , with a half-power semi-angle of $\Phi_{1/2}=60°$. The receiver adopts the same compact inclined circular PD array as that used in the simulations, consisting of eight neighboring PDs with an inclination angle of 45° and an array radius of 2 cm. Each PD has a field of view of 90°, an active area of $A_{PD}=0.49$ mm^2 , and a responsivity of $\Re=0.35$ A / W .

(2) System calibration

Prior to the experiments, the light source, PD array, and attitude sensors were jointly calibrated. Photocurrent measurements collected at known positions were employed to determine the LED parameters and overall system gain. Dark-current and zero-offset corrections were subsequently applied to each PD channel, followed by IMU calibration and magnetometer compensation. Finally, a unified timestamp was adopted to ensure synchronization between the PD and attitude measurements.

(3) Joint positioning and communication experiment

The pitch, roll, and yaw angles were fixed at 0°, and 50 samples were collected at each test point. For the fixed-plane scenario, the reference point set to $\left[3.72,3.72,0.84\right]^{\mathrm{T}}$, $\mathbf{n}_{\Pi}=\left[0,0,1\right]^{\mathrm{T}}$, respectively, and Mode B was adopted. The receiver test positions were arranged in a staggered pattern with a spacing of 0.62 m. The joint communication and positioning results are shown in Fig. 10. The estimated positions remain concentrated around the ground-truth locations, with a mean positioning error of 8.34 cm. Compared with the minimum error of 5 cm reported in Ref. [4] and the approximately 2–35 cm range reported in Ref. [11], the proposed system demonstrates competitive positioning performance. Consistent with the simulation results, the positioning error increases with the horizontal distance from the LED projection point, while similar trends are observed for noise fluctuations and the bit-error rate.

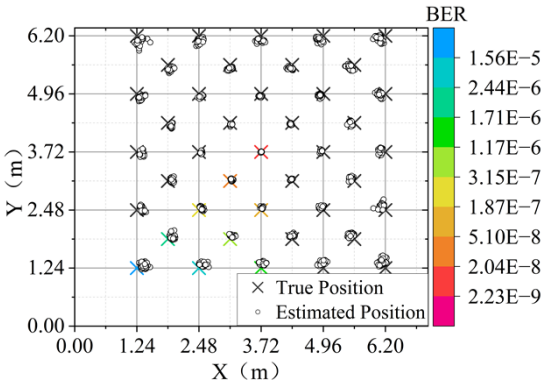


Fig. 10. Experimental results of joint positioning and communication.

(4) Three-dimensional positioning comparison under inclined conditions

For comparison with Ref. [2], three-dimensional positioning experiments were conducted under different receiver inclinations. Four test positions with a spacing of 0.5 m were arranged around the LED projection point. Receiver heights of 1.66, 1.16, and 0.66 m corresponded to vertical LED distances of 1.5, 2, and 2.5 m, respectively. The yaw angle was fixed at 0°, while the pitch and roll angles were set to 0°, 15°, and 30°. Five measurements were collected for each condition, and Mode B was employed for positioning. As shown in Fig. 11, the estimated positions remain concentrated around the ground-truth locations, yielding a mean positioning error of 5.65 cm, approximately 29% lower than that reported in Ref. [2]. The positioning error increases with propagation distance and receiver inclination, consistent with the simulation results. The residual errors are mainly attributed to modeling inaccuracies, calibration errors, and other environmental nonidealities.

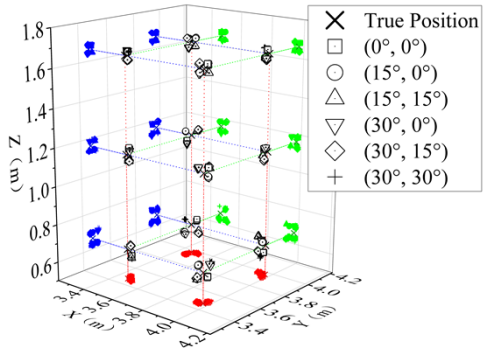


Fig. 11. Three-dimensional positioning results under inclined conditions.

5. Conclusion

This work investigates single-LED integrated visible-light communication and positioning for indoor mobile terminals with arbitrary receiver positions and orientations. A unified optical channel model and compact inclined PD array are developed, while Fisher information matrix and Cramér–Rao lower bound analyses are employed to characterize positioning performance. By incorporating IMU attitude priors, a joint position–attitude estimation algorithm is established to enhance robustness and accuracy. Prototype experiments achieve mean positioning errors of approximately 7 cm on a fixed receiving plane and 15 cm under three-dimensional free-position and free-attitude conditions, demonstrating the feasibility of single-LED communication and positioning.

Funding. National Natural Science Foundation of China (92052106, 61771385); Key Industrial Chain Innovation Team of Shaanxi Province (2024RS-CXTD-12); Key Research and Development Program of Shaanxi Province (2023-YBGY-039).

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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